Gain control of firing rate by shunting inhibition: Roles of synaptic noise and dendritic saturation

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Adjusting input–output gain is crucial for information processing by the brain. Gain control of subthreshold depolarization is commonly ascribed to increased membrane conductance caused by shunting inhibition. But contrary to its divisive effect on depolarization, shunting inhibition on its own fails to divisively modulate firing rate, apparently upsetting a critical tenet of neural models that use shunting inhibition to achieve gain control. Using a biophysically realistic neuron model, we show that divisive modulation of firing rate by shunting inhibition requires synaptic noise to smooth the relation between firing rate and somatic depolarization; although necessary, noise alone endows shunting inhibition with only a modest divisive effect on firing rate. In addition to introducing noise, synaptic input is associated with a nonlinear relation between somatic depolarization and excitation because of dendritic saturation; this nonlinearity dramatically enhances divisive modulation of firing rate by shunting inhibition under noisy conditions. Thus, shunting inhibition can act as a mechanism for firing rate gain control, but its modulatory effects (which include both divisive and subtractive components) are fully explained only when both synaptic noise and dendritic saturation are taken into account.

Gain control refers to modulation of a neuron's responsiveness to input; it is critically important for normal sensory, cognitive, and motor functions (1–5). Gain control is achieved through a divisive process and is observed as a change in the slope of the input–output curve. A longstanding and widely cited candidate mechanism for gain control is shunting inhibition (6, 7) such as that mediated by GABA-gated Cl^- conductance, which decreases input resistance with little direct effect on membrane potential (V_m). Whereas hyperpolarization has a subtractive effect on the depolarization caused by concurrent excitation, shunting has a divisive effect (Fig. 1*A*), thereby modulating gain of synaptic depolarization.

However, neurons communicate with each other through spike trains. One must, therefore, consider how V_m translates into firing rate. Contrary to initial intuition, modeling studies (8–10) supported by *in vitro* electrophysiological data (11, 12) concluded that shunting inhibition modulates firing rate in a purely subtractive manner. However, electrophysiological studies that analyzed responses in the presence of background synaptic input (13–15) provide circumstantial evidence supporting a role for shunting inhibition in gain control. These dichotomous findings suggest that properties of synaptic input such as its noisiness or its effect on membrane conductance (16, 17), which were neglected in past modeling, may explain the discrepancy regarding the role of shunting inhibition in gain control. Only recently did Chance *et al.* (18) independently demonstrate that background synaptic noise was key for achieving firing rate gain control by shunting.

This study was therefore undertaken to examine whether shunting inhibition can divisively modulate firing rate when a neuron is bombarded by synaptic input comparable to that recorded in the intact brain. The data demonstrate that shunting inhibition can indeed mediate firing rate gain control so long as both synaptic noise and dendritic saturation are taken into account.

Methods

Compartmental Model. A compartmental model was constructed from a layer V neocortical pyramidal neuron described in ref. 19. An axon morphologically identical to that in ref. 20 was attached to the soma. A passive leak conductance reversing at −70 mV was inserted at 0.02 $mS \cdot cm^{-2}$ in all compartments except axonal nodes, where it was 10-fold higher, giving an input resistance of 59 MΩ and membrane time constant of ≈40 ms. Axial resistivity was set to 200 Ωcm, and membrane capacitance was set to 1 $\mu F \cdot cm^{-2}$ except in axon internodes, where it was 0.04 $\mu F \cdot cm^{-2}$. Voltage-dependent Na^+ and K^+ conductances [Hodgkin–Huxley (HH) channels] were inserted at different densities depending on the compartment (Na^+, K^+ in $mS \cdot cm^{-2}$): soma and primary dendrites, 48, 40; distal dendrites (≥ secondary), 6, 5; initial segment, axon hillock, and nodes, 240, 200; axon internodes, 6, 5. Kinetics of these currents were taken from ref. 21.

Synaptic Input. Synaptic input was modeled as a nonsaturating conductance with a dual exponential time course of the form $[1-\exp(-t/\tau_{rise})]\exp(-t/\tau_{decay})$. For excitatory input, $\tau_{rise} = 0.2$ ms, $\tau_{decay} = 2$ ms, $g_{max} = 1$ nS, and $E_{rev} = 0$ mV; for inhibitory input, $\tau_{rise} = 0.8$ ms, $\tau_{decay} = 8$ ms, $g_{max} = 0.5$ nS, and $E_{rev} = -70$ mV (10, 16, 17, 22–24). Two sets of eight inhibitory synapses were positioned randomly in the perisomatic region. One set of 24 excitatory synapses was positioned on the basal dendrites, and another set was positioned on the apical tuft. The number of synapses per set reflects a combination of the number of contacts per presynaptic cell (23, 25) and the coordination of presynaptic activity (17, 26–28) and was adjusted to generate V_m fluctuations of appropriate amplitude (17). Each set was driven by an independent Poisson process. The total number of synapses is estimated to be 24,000 excitatory (26 synapses per 100 μm^2 of dendritic surface area; ref. 29) and 4,800 inhibitory (one-fifth the number of excitatory synapses; ref. 17); the 48 excitatory and 16 inhibitory model synapses were therefore driven, respectively, at 500 times and 300 times the event frequency of real synapses, essentially replacing spatial summation with temporal summation. Validity of this assumption was tested in a model cell with a 10-fold increase in the number of sets of synapses and equivalent decrease in frequency of events per synapse; this more complex model reproduced all critical findings reported in Figs. 1 and 2. Data from the more computationally efficient model are reported throughout. Reported frequencies refer to events per synapse after correction for synapse number.

Nonsynaptic Stimulation. Constant current (I_{clamp}) was injected into the soma to simulate the typical electrophysiological exper-

Abbreviations: f_{exc}, rate of excitatory synaptic events; f_{inh}, rate of inhibitory synaptic events; f_{out}, output firing rate; g_{exc}, excitatory conductance; g_{inh}, inhibitory conductance; HH, Hodgkin–Huxley; I_{clamp}, injected current; V_m, membrane potential.

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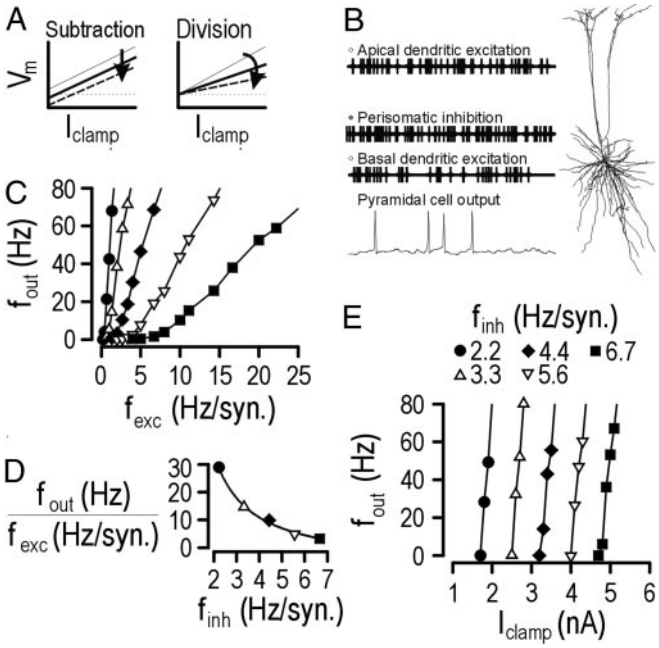


Fig. 1. Effect of shunting inhibition on firing rate depends on stimulus conditions. (*A*) Hyperpolarization has a subtractive effect on subthreshold depolarization, whereas shunting has a divisive effect. (*B*) Image of the model cell with a sample of Poisson-distributed excitatory and inhibitory synaptic inputs and the resultant output. (*C*) Gain of the relationship between output firing (f_{out}) and excitatory synaptic input (f_{exc}) was reduced by increasing inhibitory synaptic input (f_{inh}). Although this qualifies as a divisive effect, a subtractive effect is also evident as a shift in the curves, i.e., position along abscissa where f_{out} rises above 0 Hz increases as f_{inh} increases. (*D*) Reduction in slope of f_{out}-f_{exc} curves was nonlinearly related to f_{inh}. Slope was determined from linear regressions of data in *C* using all values of $f_{out} > 5$ Hz. (*E*) Effect of shunting inhibition on f_{out} was purely subtractive when excitatory and inhibitory synaptic input were replaced with constant current injection (I_{clamp}) and constant conductance change (g_{inh}) applied to the soma. Values of g_{inh} correspond to the sum of time-averaged conductances of all inhibitory synapses for f_{inh} levels illustrated in *C* and *D*.

iment. In some cases, to remove noise but still mimic the conductance-altering properties of synaptic input, fluctuating synaptic conductances were replaced with constant excitatory conductance (g_{exc}) and/or inhibitory conductance (g_{inh}) simulated as either a single-point process in the soma or as multiple-point processes at synaptic sites. Reported values of g_{exc} and g_{inh} refer to total excitatory or inhibitory conductance, based on the time-averaged synaptic conductances scaled by the number of synapses. Contrariwise, to recreate V_m fluctuations caused by synaptic input without the associated conductance change, noisy current was generated by an Ornstein–Uhlenbeck process and injected into the soma. We used a process like that described in ref. 30 where current fluctuates according to $dI/dt = -1/\tau\,(I - I_{avg}) + \sigma\sqrt{}dt\,N(0, 1)$, where $N(0, 1)$ is a number drawn from a Gaussian distribution with average 0 and unit variance with a scaling factor σ, which adjusts noise amplitude. The power spectrum of V_m generated with $\tau = 5$ ms reasonably approximates that generated by synaptic input.

All simulations were conducted with NEURON 4.3.1 by using a 10-μs time step and simulated temperature of 36°C. Segmentation of compartments was defined by using the d_lamba criterion in NEURON (31), such that grid points were spaced no further than 10% of the length constant at 100 Hz. Each datum is based on 10 s of simulated time for output firing rate (f_{out}) values and 1 s for V_m values.

Results

Morphology of the model cell and a representative response to synaptic input are shown in Fig. 1*B*. Under conditions approximating *in vivo* background synaptic input [rate of excitatory synaptic events (f_{exc}) = 1.0 Hz/synapse and rate of inhibitory synaptic events (f_{inh}) = 5.6 Hz/synapse] (17), our model neuron was depolarized to −60.7 mV, standard deviation of V_m was 2.5 mV, and R_{in} was reduced by 89%. This is comparable to previously reported values (16, 17) indicating that model parameters were set within reasonable bounds.

A critical finding of this study was that shunting inhibition mediates firing rate gain control when the model cell was bombarded by realistically simulated synaptic input. Slope of the relationship between output firing rate and frequency of excitatory input was clearly reduced as the frequency of inhibitory input was increased (Fig. 1 *C* and *D*). However, the effect of shunting was not purely divisive; the rightward shift in the f_{out}-f_{exc} curves points to a subtractive component. Equivalent divisive/subtractive modulation was observed when the number of simulated synapses was increased (see *Methods*) and in two different model neurons tested (data not shown). When spike frequency adaptation was incorporated into the model neuron by addition of an M-type K^+ current, the divisive effect was still evident in both the preadapted and adapted firing rates (data not shown).

This result is inconsistent with previous modeling studies that concluded shunting inhibition had a purely subtractive effect on f_{out} (8–10). The main difference between models was the manner in which synaptic input was simulated. We therefore tested our model neuron in the absence of synaptic noise by generating excitation with constant somatic I_{clamp} and shunting with constant somatic g_{inh}. Under these conditions, shunting inhibition had a purely subtractive effect on f_{out} (Fig. 1*E*), implicating properties of the synaptic stimulation as key factors for achieving division.

Neuronal Behavior in the Presence of Synaptic Input. How does synaptic input alter neuronal behavior to allow divisive modulation of f_{out} by shunting inhibition? To first investigate the impact of synaptic input without the added complexity of spike generation, we removed HH channels from all compartments except distal dendrites. This reduced HH condition prevented spiking while still encouraging propagation of distally evoked depolarization into the soma. Fig. 2*A* shows the noisy fluctuations in somatic V_m produced by random synaptic input. Increased shunting had two effects: reduction of average somatic depolarization caused by excitatory input (Fig. 2*B*) and reduction of V_m fluctuation amplitude (Fig. 2*C*). Notably, standard deviation of V_m did not reduce to 0 mV, showing that fluctuations remain even in the presence of strong shunting. The near-horizontal regression lines in Fig. 2*C* indicate that V_m fluctuation amplitude was relatively unaffected by the strength of excitation, consistent with previously reported data (17, 18, 30).

Because of its fluctuations, somatic V_m occasionally crosses spike threshold despite average V_m remaining below threshold (Fig. 2*A*). Therefore, probability of crossing threshold depends jointly on average V_m (i.e., distance from threshold) and amplitude of V_m fluctuations. The consequences of this for spiking are seen in Fig. 2*D* after reinsertion of HH channels. Constant I_{clamp} gives rise to an abrupt increase in f_{out} at rheobase (i.e., minimum constant I_{clamp} eliciting spikes), where probability of spiking jumps from 0 to 1. By contrast, f_{out} increases more gradually in response to noisy input because the V_m fluctuations allow for probabilistic or noise-driven spiking before average depolarization reaches threshold. This amounts to smoothing the threshold nonlinearity, allowing f_{out} to encode the average underlying V_m even while average V_m remains below threshold (32–35). Given

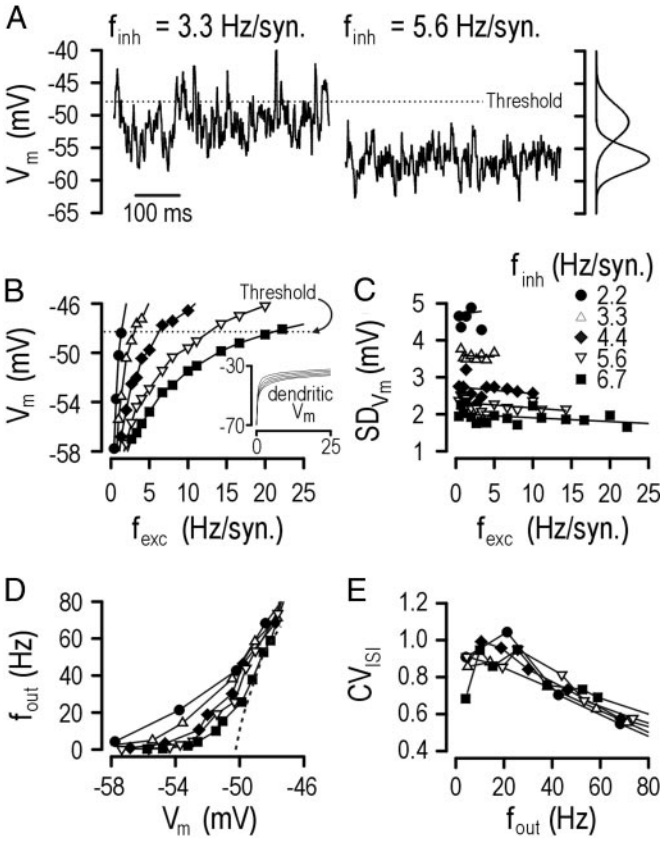


Fig. 2. Synaptic input introduces significant noise into the postsynaptic neuron, with ramifications for spike generation. (*A*) Sample of somatic V_m fluctuations in model neuron excited by random synaptic input (f_{exc} = 2 Hz/synapse) for two levels of f_{inh}. HH channels were removed from all compartments except distal dendrites (reduced HH condition; see text) to prevent spiking. (*B*) Increasing inhibition divisively reduced the average somatic depolarization caused by synaptic excitation. The somatic V_m-f_{exc} curves are nonlinear because of dendritic saturation; *Inset* shows V_m averaged across four different dendritic locations (see also Fig. 4). (*C*) Inhibition also reduced the standard deviation of V_m (SD_{Vm}), although this reduction appeared to asymptote near 2 mV. Excitation appeared to have relatively little influence on SD_{Vm}, as evidenced by the near-horizontal regression lines (see text). (*D*) This graph was generated by plotting f_{out} in the normal HH condition (from Fig. 1*C*) against somatic V_m for equivalent stimulation in the reduced HH condition (from *B*). It illustrates that f_{out} depends on both average V_m and SD_{Vm} (which is modulated by shunting inhibition; see *C*). Furthermore, fluctuations in V_m allow spiking to occur despite average V_m being below threshold (i.e., −48 mV), which we refer to as smoothing the threshold nonlinearity; compare with dashed line showing response to constant I_{clamp}, which falls abruptly to f_{out} = 0 Hz. (*E*) Spiking caused by V_m fluctuations is associated with high variability [quantified as the coefficient of variation of the interspike interval (ISI), CV_{ISI}, calculated as the standard deviation of the ISI divided by the average ISI], which decreases as the average V_m approaches and rises above spike threshold, which occurs at $\approx f_{out}$ = 50 Hz (see *D*). Given f_{out} values observed *in vivo*, neurons often operate in this subrheobasic regime where ''probabilistic'' spiking is driven by noise.

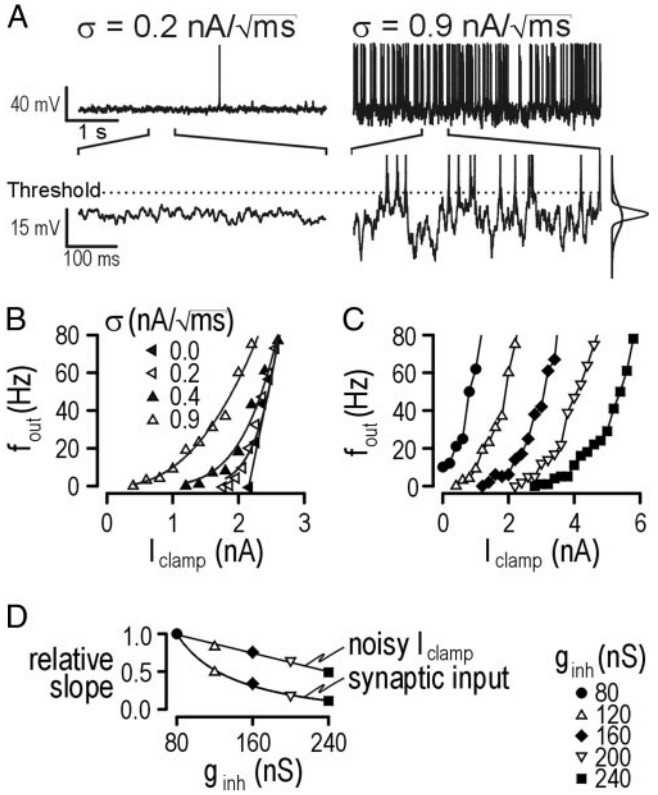


Fig. 3. Effect of V_m fluctuations on spike generation (smoothing the threshold nonlinearity) is critical for allowing divisive modulation of firing rate by shunting inhibition but cannot fully account for the divisive effect seen with synaptic input. (*A*) Sample responses of the model neuron stimulated by noisy I_{clamp}, where σ determines noise amplitude. Effect of σ on the distribution of V_m is shown on the right; values of V_m during spikes were excluded from the calculation. (*B*) V_m fluctuations generated by injection of noisy I_{clamp} smoothed the increase of f_{out}. Reducing V_m fluctuations by reducing input noise (decreasing σ) altered f_{out}-I_{clamp} curves in the same manner as reducing V_m fluctuations by increasing inhibition altered f_{out}-V_m curves in Fig. 2*D*. This demonstrates that modulating V_m fluctuation amplitude does not mediate a divisive effect on f_{out}; on the contrary, this modulation subtractively influences f_{out}. (*C*) With excitation by noisy I_{clamp} and inhibition by constant g_{inh} applied to the soma, a modest divisive effect on f_{out} was observed, being greatest when σ was high (σ = 0.9 nA/$\sqrt{}$ms for illustrated curves) and degenerating into a purely subtractive effect as σ was reduced, with σ = 0 nA/$\sqrt{}$ ms corresponding to constant I_{clamp} conditions in Fig. 1*E*. (*D*) Under noisy I_{clamp} and constant g_{inh} stimulus conditions, the relative change in slope (calculated from linear regressions of data in *C* using $f_{out} < 40$ Hz) with increasing g_{inh} was far less than observed under synaptic conditions (data from Fig. 1*D*).

that smoothing allows f_{out} to more closely parallel V_m, divisive modulation of V_m may be predicted to manifest as divisive modulation of f_{out} under noisy conditions.

Spiking recorded *in vivo* is normally highly irregular (36, 37). Fig. 2*E* shows that spiking caused by V_m fluctuations (which we refer to as probabilistic spiking) is irregular compared with spiking generated by raising average V_m above threshold (27, 38–41). Under *in vivo* conditions, neurons typically operate in the lower range of f_{out}, in a regime where spiking is driven by noise (42, 43). We therefore paid particular attention to this regime of probabilistic spiking.

Role of Synaptic Noise. Having identified noisy fluctuations in V_m as important for determining spiking elicited by synaptic input, an obvious question was whether noise alone is sufficient to allow shunting inhibition to divisively modulate f_{out}. To isolate the influence of noise, V_m fluctuations with amplitude and frequency content similar to that generated by synaptic input were reconstituted in the model neuron by directly injecting noisy current into the soma (Fig. 3*A*). As with synaptic input (Fig. 2*D*), V_m fluctuations generated by noisy I_{clamp} smoothed the threshold nonlinearity in proportion to the intensity of the noise (Fig. 3*B*).

Under noisy conditions, shunting had some divisive effect on f_{out} (Fig. 3*C*) in contrast to the purely subtractive effect observed with constant I_{clamp} (Fig. 1*E*). However, Fig. 3*C* shows that the reduction in slope of f_{out}-I_{clamp} curves was restricted to probabilistic firing (i.e., lower values of f_{out}), with curves becoming parallel at $f_{out} > 40$ Hz. Furthermore, based on responses with

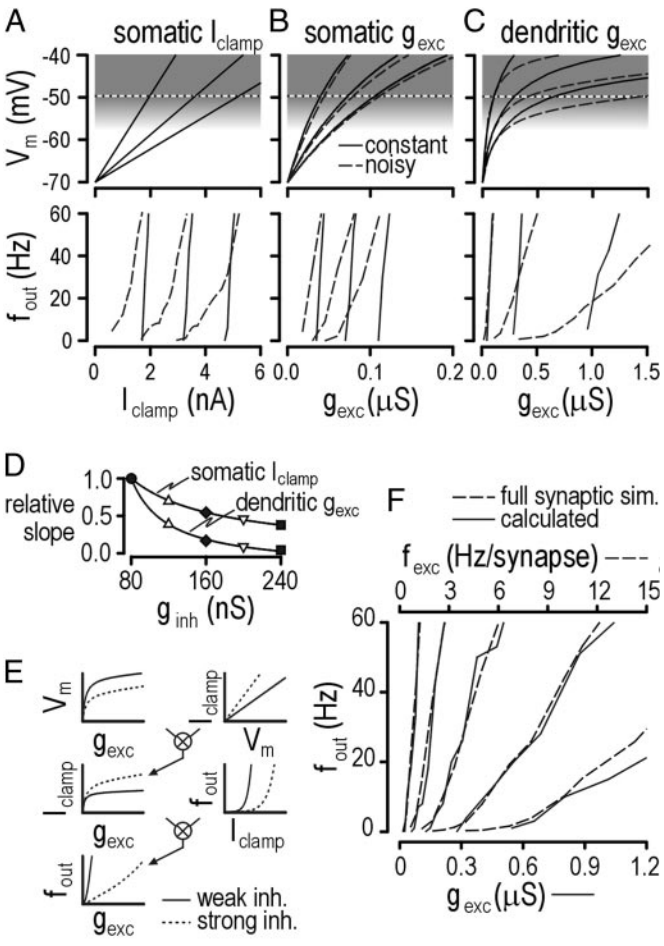


Fig. 4. The nonlinearity caused by dendritic saturation enhances divisive modulation of f_{out} by shunting inhibition. Gray shading in *A–C* represents the probability of spiking within an arbitrary time interval: darkest gray → probability = 1; white → probability = 0. The gradient of gray shading below threshold (dotted white line) results from threshold smoothing (see Fig. 2*D*) and exists only in the presence of noise; it is referred to as the region of probabilistic firing. (*A*) In the reduced HH condition, somatic V_m is linearly related to somatic I_{clamp} (*Upper*). The divisive effect of shunting on these linear curves manifests as a subtractive effect on f_{out}-I_{clamp} curves (*Lower*) under noise-free conditions (solid lines) and becomes only mildly divisive in noisy conditions (dashed lines). (*B*) The relation between somatic V_m and somatic g_{exc} is nearly linear (*Upper*) and gives rise to f_{out}-g_{exc} curves (*Lower*), very similar to those observed in *A*. (*C*) In contrast, excitation by dendritic g_{exc} produces a highly nonlinear relation between somatic V_m and g_{exc} (*Upper*). Note the difference in scale in the abscissa between *B* and *C*. The difference between solid and dashed lines (constant and noisy stimulation, respectively) reflects the differential electrotonic attenuation of signals depending on their frequency content. The nonlinearity results from dendritic saturation (see text). Because of the nonlinearity, the divisive effect of shunting dramatically alters the trajectory of those curves through the region of probabilistic firing, which manifests as enhanced divisive modulation of the f_{out}-g_{exc} curves (*Lower*). (*D*) The relative change in trajectory with increasing g_{inh} is summarized for somatic I_{clamp} and dendritic g_{exc} stimulus conditions. The difference between curves predicts the amount of divisive modulation of f_{out} before and after taking into account the effects of dendritic saturation and should be compared with data in Fig. 3*D*. (*E*) To test whether threshold smoothing by noise together with the nonlinearity caused by dendritic saturation can fully explain the divisive effects of shunting observed under synaptic conditions, it is necessary to combine these independently measured effects to see how, together, they influence f_{out}. The schematic shown here illustrates how f_{out}-g_{exc} curves in *F* were generated; example curves corresponding to weakest and strongest inhibition (solid and dotted lines, respectively) are shown. V_m-g_{exc} curves (corresponding to dashed lines in *C Upper*) were combined with I_{clamp}-V_m curves (corresponding to mirror image of *A Upper*) to produce I_{clamp}-g_{exc} curves, allowing one to infer the value of somatic I_{clamp} necessary to produce equivalent somatic depolarization as that produced by noisy g_{exc} in the dendrites; this introduces the nonlinearity caused by dendritic saturation. These I_{clamp}-g_{exc} curves were then combined with f_{out}-I_{clamp} curves with smoothed threshold (corresponding to dashed lines in *A Lower*, showing response to $\sigma = 0.4$ nA/$\sqrt{}$ ms noise level, which best reproduces the values of standard deviation of V_m reported in Fig. 2*C*) to produce the f_{out}-g_{exc} curves shown in *F* as solid lines. This introduces the effect of threshold smoothing caused by noise. Thus, calculated f_{out}-g_{exc} curves are directly comparable to measured f_{out}-f_{exc} curves but represent the spiking response to artificial noise (i.e., noisy somatic I_{clamp}) with the values of I_{clamp} converted to values of g_{exc} by nonlinear scaling. (*F*) The calculated f_{out}-g_{exc} curves (solid lines) closely approximate the f_{out}-f_{exc} curves generated by synaptic input (dashed lines), which correspond to the data in Fig. 1*C*.

$f_{out} < 40$ Hz, the reduction in slope was modest compared with that observed under synaptic conditions for the same amount of shunting (Fig. 3*D*). Thus, the divisive effect of shunting inhibition under synaptic conditions is not fully explained simply by adding noise and, except under certain circumstances (e.g., with precisely balanced excitation and inhibition), the subtractive component predominates (Fig. 6, which is published as supporting information on the PNAS web site, www.pnas.org). What other aspect of synaptic input is responsible for the remaining division?

Role of Dendritic Saturation. We have argued that noisy V_m fluctuations allow f_{out} to more accurately encode average V_m by smoothing the threshold nonlinearity. However, the relationship between excitatory input and somatic depolarization depends on the stimulus. Somatic depolarization is linearly related to somatic I_{clamp} (Fig. 4*A Upper*), and the relationship becomes only modestly sublinear for excitation by somatic g_{exc} (Fig. 4*B Upper*). In contrast, somatic depolarization is nonlinearly related to dendritic g_{exc} (Fig. 4*C Upper*), which is particularly important given that most excitatory synapses occur on dendrites. The strong nonlinearity results from a complex interplay between multiple factors (see *Discussion*), which we simply refer to as dendritic saturation.

Analysis of the effects of shunting inhibition on somatic V_m-excitation curves (Fig. 4 *A Upper–C Upper*) allows one to roughly predict the subtractive effect of shunting inhibition on f_{out} for comparison to f_{out}-excitation curves (Fig. 4 *A Lower–C Lower*). Without noise, the subtractive effect of shunting on f_{out} is predicted by where V_m-excitation curves cross spike threshold. Introducing noise reduces the effective threshold for spiking such that the subtractive effect on f_{out} can be predicted from where V_m-excitation curves enter the region of probabilistic spiking. The subtractive effect is therefore reduced in noisy conditions.

Insight into the divisive effect of shunting can be similarly gained by considering alteration in the trajectory of V_m-excitation curves through the region of probabilistic spiking. Acting on the linear V_m-somatic I_{clamp} curves and near-linear V_m-somatic g_{exc} curves (Fig. 4 *A* and *B*, respectively), shunting causes a rather modest change in trajectory compared with the strong change observed when shunting acts on the nonlinear V_m-dendritic g_{exc} curves (Fig. 4*C*). The relative changes in trajectory for V_m-somatic I_{clamp} and V_m-dendritic g_{exc} curves are summarized in Fig. 4*D* and correctly predict the enhanced divisive effect of shunting on f_{out} (compare with Fig. 3*D*), depending on stimulus conditions. The critical observation is that the nonlinearity caused by dendritic saturation enhances the divisive effect of shunting on f_{out} much more than it enhances the subtractive effect because of the shape of the nonlinearity relative to the region of probabilistic firing.

Thus the combined effects of two factors, threshold smoothing by synaptic noise and the nonlinearity caused by dendritic

saturation, appear to explain how shunting inhibition modulates f_{out}. We sought to quantitatively test the adequacy of this explanation by reconstituting a family of f_{out}-g_{exc} curves representing different levels of inhibition for comparison to the f_{out}-f_{exc} curves in Fig. 1*C*. In other words, these f_{out}-g_{exc} curves do not represent results of additional simulations but rather are calculated from earlier simulations aimed at independently measuring the threshold smoothing caused by noise (Fig. 3) and the nonlinearity introduced by dendritic saturation (Fig. 4*C*). The manner in which the multiple different relationships described earlier in the paper were combined to generate f_{out}-g_{exc} curves directly comparable to f_{out}-f_{exc} curves is explained in Fig. 4*E*. The calculated f_{out}-g_{exc} curves exhibit a pattern of mixed divisive and subtractive change (Fig. 4*F*, solid lines) closely resembling that seen in response to synaptic excitation (dashed lines; data from Fig. 1*C*). By comparison, applying the modest nonlinearity associated with somatic g_{exc} (which circumvents the effects of dendritic saturation; see Fig. 4*B*) could not accurately reproduce the data in Fig. 1 *C* and *D* (Fig. 6). Thus, synaptic noise and dendritic saturation both influence how synaptic input is transduced into firing rate, and both must be taken into account to fully explain the outcome of shunting inhibition on firing rate.

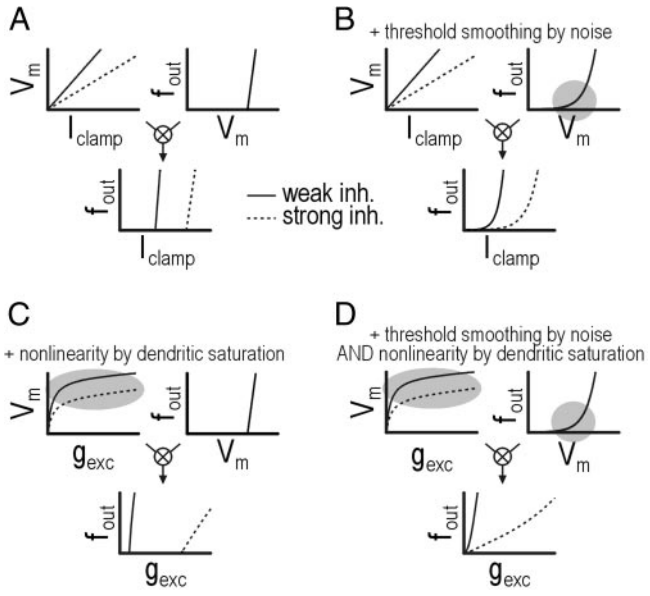


Fig. 5. Effect of shunting inhibition on f_{out} under different conditions. (*A*) Without noise or dendritic saturation, the divisive effect of shunting on V_m-I_{clamp} curves manifests as a subtractive effect on f_{out}-I_{clamp} curves. (*B*) Smoothing the threshold nonlinearity in the f_{out}-V_m curve by noise (highlighted with gray) endows shunting with some capacity to divisively modulate the f_{out}-I_{clamp} curve. (*C*) Excitatory conductance in the dendrites results in dendritic saturation, which introduces a strong nonlinearity into relation between somatic V_m and g_{exc} (highlighted with gray), but, in the absence of threshold smoothing by noise, the divisive effect of shunting on the nonlinear V_m-g_{exc} curves manifests as a subtractive effect on the f_{out}-g_{exc} curves. (*D*) When both threshold smoothing by noise and the nonlinearity caused by dendritic saturation are taken into account, the divisive action of shunting on the nonlinear V_m-g_{exc} curves manifests as predominantly divisive modulation of the f_{out}-g_{exc} curves, accurately recreating the modulation observed under synaptic conditions (Fig. 1*C*).

Discussion

Data presented here indicate that shunting inhibition can indeed divisively modulate firing rate, but that this ability relies on two effects of synaptic input: the noise it causes in somatic V_m and the saturation it causes in dendrites. In the absence of those effects, shunting has a purely subtractive effect on f_{out} (Fig. 5*A*). Threshold smoothing by noise allows a modest divisive reduction of f_{out} by shunting (Fig. 5*B*). On its own, the nonlinearity caused by dendritic saturation does not allow a divisive effect of shunting on f_{out} (Fig. 5*C*). However, when combined with threshold smoothing by noise, the effects of dendritic saturation serve to dramatically enhance divisive modulation of f_{out} by shunting inhibition relative to the residual subtractive effect (Fig. 5*D*).

The results of this study explain the enigmatic link between shunting inhibition and gain control of firing rate. The discrepancy between our results and those of past modeling studies, which concluded that shunting inhibition could not mediate firing rate gain control (8–10), is accounted for by the presence or absence of synaptic noise in the model. This is supported by a recent modeling study that has also implicated noise in determining the effect of shunting inhibition (44) and is consistent with data in ref. 17. Neurons *in vivo* experience significant levels of synaptic noise (17, 33, 34, 40) and therefore exist under conditions where shunting is likely to act divisively on f_{out}.

How does noise allow introduction of a divisive effect? Studies on stochastic resonance have shown that noise can enhance the detection of weak signals (45, 46). This increased sensitivity reflects altered transduction of graded input into spikes: Noise smoothes the threshold nonlinearity, converting the probability of spiking in response to a given input from a step function into a sigmoidal one (34), with the result that firing rate more smoothly encodes the underlying depolarization (32, 35, 47). This smoothing allows for the contrast invariance of orientation tuning in the visual cortex (33) and, as shown in the present study and by Chance *et al.* (18), it also allows firing rate to reflect the divisive influence of shunting on V_m.

One benefit of our approach was that we made no assumptions *a priori* as to what factors might be involved in allowing gain control by shunting. Instead, we dissected out those factors and tested whether and to what degree shunting inhibition could divisively modulate f_{out} when factors were taken into account independently or in combination. This approach revealed that a second factor, the nonlinearity caused by dendritic saturation, combined with synaptic noise to determine the final outcome of shunting inhibition on f_{out}. This nonlinearity stems from the fact that electrotonic attenuation causes only a fraction of the excitation produced in dendrites to make it to the soma to mediate somatic depolarization (48, 49); therefore, strong dendritic excitation is necessary to achieve even moderate somatic excitation. But the necessity for strong dendritic excitation results in dendritic saturation; in other words, dendrites becomes depolarized to the extent that the driving force associated with excitatory synaptic conductance becomes significantly reduced (50). Consequently, excitatory conductance becomes progressively less capable of generating excitatory current as overall excitation (and, by extension, dendritic saturation) increases. Compounding this, the requisite high level of synaptic input significantly alters the electrotonic structure of the neuron and exacerbates the attenuation (16). The end result is a highly nonlinear relation between somatic V_m and dendritic g_{exc}, which, because of the shape of that nonlinearity relative to the region of probabilistic firing, enhances the divisive modulation of f_{out} by shunting inhibition.

During revisions of this paper, Chance *et al.* (18) independently demonstrated the role of noise in firing rate gain control using dynamic clamp. Our data are consistent with their findings but argue that in addition to synaptic noise, dendritic saturation, which they did not consider, enhances shunting inhibition's divisive effect on f_{out} relative to its subtractive effect and thereby

diminishes the necessity of some of the assumptions they made to demonstrate gain control (see Fig. 6).

The present study has shown that the ability of shunting inhibition to divisively modulate firing rate depends critically on synaptic noise. Moreover, divisive modulation of firing rate is greatly enhanced relative to the residual subtractive effect by the nonlinearity caused by dendritic saturation. The consequences of noise and dendritic saturation, both of which result from the strong background synaptic input characteristic of *in vivo* conditions, demonstrate the impact of network activity on the computational capabilities of individual neurons.

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